Fig. 2: 2014 FIFA World Cup: Juliano Pinto, paralysed from his chest down is making his first kick in the opening ceremony with the help of exoskeletons

Fig. 3: (a) 'Hardiman'—the first exoskeleton by General Electric, (b) Tom Cruise with his exosuit in 'Edge of Tomorrow,' and (c) Matt Damon with his exoskeleton in 'Elysium'

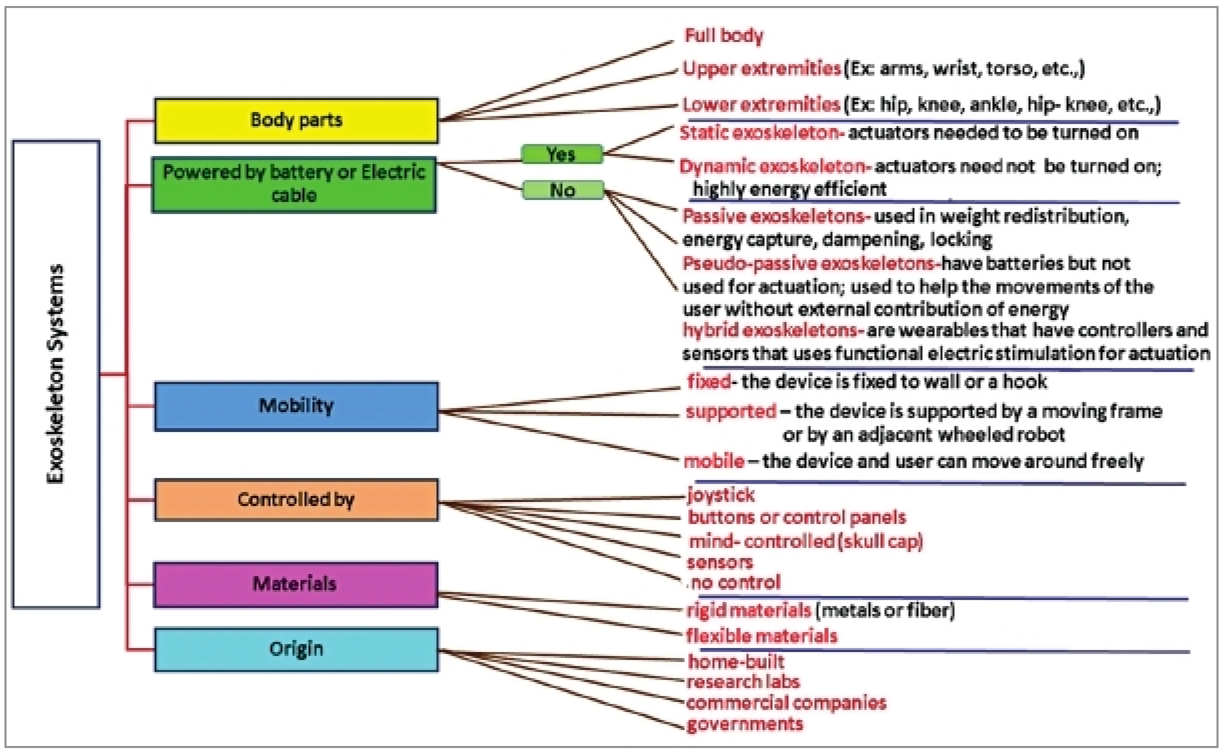

Fig. 4: Classification on exoskeleton systems

of untethered devices. So, the recent past has seen diverse applications of exoskeletons in the healthcare industry due to their exceptional ability to rehabilitate patients with physical or neurological disorders, thus enhancing their performance capability and functionality.

These devices initially entered the medical industry to assist healthcare professionals in lifting and carrying patients. Later, researchers transferred their focus to patients as well. The first working medical exoskeleton was built in 1972 by Mihajlo Pupin Institute in Belgrade, Serbia. Exoskeleton systems in the healthcare industry are classified into different categories, as shown in Fig. 4.

However, in a gait training, the classification is narrowed down into two categories—end-effectors: devices attached at the end of the exoskeleton, and powered exoskeletons: wearable robotics powered by a system of electric motors, pneumatics, levers, hydraulics, or a combination of technologies that allow limb movements. For patients suffering from neurological conditions, particularly leading to gait disorders, exoskeletons are becoming a powerful tool in their rehabilitation process. Thus, the physiotherapy and mobility reforms of the healthcare sector are getting transformed due to exoskeletons.

Need for exoskeletons

Earlier, restoration processes that were used to correct the walking disabilities involved locomotor training in which the mechanical devices used allowed individuals with gait disorders to stand or walk on a bodyweight supported treadmill. The placement of feet could be manual or using a robotic system.

However, such methods had limited success due to the high cost related to ambulation, difficulty in obtaining the required skills and strength, and the incapability to continue the therapy at home. Exoskeletons, especially the powered types, are wearable robots that enhance the strength, endurance, and mobility that increase the independence of the user.

Exoskeletons are highly flexible and can be used for performing a wide range of limb movements like walking, climbing stairs, sitting, and standing up. These, in turn, decrease the number of rehabilitation tools required, the number of therapists required for various tools involved, and hence reduce the cost of the rehabilitation process.

Exoskeletons are also available for patients for their upper limb(s). By using exoskeletons, patients can go through intensive all-day functional therapy during their daily life activities. Also, several researches have proved that exoskeletons facilitate the safety of the user while walking. Other advantages include a reduction in spasticity and pain.

Exoskeletons for gait and balance

Walking is a precious gift given by God, and we often take it for granted. It is a complex task that requires the